Valve core for a tyre valve, seat ring, and methods of use and manufacture — Prosecution Assessment

Likelihood of grant	6 / 10	No saved reference discloses the claimed valve core, but there is a credible obviousness combination, the only test data are internally inconsistent, the tested construction's geometry is not described consistently, and the disclosed axial disc-spring form escapes every claim. Scored on the application as reviewed in round 1; fixes made in the fix pass after review are not re-scored.

Deficiencies and prior art the application will face

#: 1 · **Deficiency or prior-art reference:** The Background ([0009]) describes a publication number that the dossier records as a fictional test reference. It was never fetched; the real document with that number is unrelated. It misstates the prior art and creates a US duty-of-candour problem. · **Type:** Formal · **Claims affected:** none (description [0009], [0015]) · **Severity:** high · **Source:** F-28-1 · **How to respond:** Delete [0009] and adjust [0015]; add nothing unless a real, saved document replaces it. · **Status:** open, routed to fix pass

#: 2 · **Deficiency or prior-art reference:** Same citation: the public record for that number is an antenna application, so the attributed content is wrong (EPO Rule 42(1)(b); US admission and IDS problem). · **Type:** Formal · **Claims affected:** none (description [0009]) · **Severity:** high · **Source:** F-30-2 · **How to respond:** As row 1; one deletion answers both. · **Status:** open, routed to fix pass

#: 3 · **Deficiency or prior-art reference:** Claims form: 45 of 86 claims are multiple dependent and most rest on other multiple dependent claims, which the USPTO does not accept; the US examiner will not examine them until the set is rebuilt. · **Type:** Formal · **Claims affected:** 13, 15–16, 18–20, 25, 30, 33, 35–38, 43–44, 46–50, 54–60, 65, 67–69, 74–77, 84, 86 · **Severity:** high · **Source:** F-37-3 · **How to respond:** File a US claim set with single dependencies on the chains that matter; no scope is lost. · **Status:** open, routed to fix pass

#: 4 · **Deficiency or prior-art reference:** Example 1, the only test data, cannot have run as recorded: forty 24-hour hold cycles in a nine-day test (12–20 August 2026). An opponent will call the data unreliable. · **Type:** Enablement · **Claims affected:** 1, 2, 3 · **Severity:** high · **Source:** F-32-4 · **How to respond:** The inventor checks the laboratory record and states the true dates, hold period and cycle counts; the text is corrected to the record only. · **Status:** open, routed to fix pass

#: 5 · **Deficiency or prior-art reference:** Same defect found by the second reviewer: dates, protocol and cycle counts of Example 1 are inconsistent. · **Type:** Enablement · **Claims affected:** 1, 2, 3 · **Severity:** high · **Source:** F-36-2 · **How to respond:** As row 4. · **Status:** open, routed to fix pass

#: 6 · **Deficiency or prior-art reference:** The tested (principal) construction is not described coherently: which end of the seat ring the purge shoulder loads, what reacts the load, how an outward-moving wall opens a gap against the bore it is pressed into, and where the purge air enters and leaves. · **Type:** Enablement · **Claims affected:** 1, 2, 3, 16–19, 22, 24, 45, 58, 65 · **Severity:** high · **Source:** F-32-6 · **How to respond:** Add one inventor-confirmed paragraph (and figure detail) from the prototype stating orientation, load path, closing contact at rest, how the gap opens, and the purge-flow route. · **Status:** open, routed to fix pass

#: 7 · **Deficiency or prior-art reference:** Same geometry problem, focused on the axial load path: [0115], [0169], FIG. 1 and FIG. 2 instructions disagree, and the convolution is said both to flatten and to bow more under load. · **Type:** Enablement · **Claims affected:** 1, 2, 3, 21, 22, 62 · **Severity:** high · **Source:** F-28-5 · **How to respond:** As row 6; conform [0115], [0136], [0239], FIGS. 1–2 and claim 22 to the prototype. · **Status:** addressed in fix pass

(not re-reviewed)

#: 8 · **Deficiency or prior-art reference:** Same geometry problem: the loaded end and the purge-air inlet past the interference fit are not stated consistently. · **Type:** Enablement · **Claims affected:** 1, 2, 16, 17 · **Severity:** high · **Source:** F-32-17 · **How to respond:** As row 6. · **Status:** addressed in fix pass (not re-reviewed)

#: 9 · **Deficiency or prior-art reference:** The text says outward movement of the ring wall opens a gap to the bore in which the ring is interference-fitted, without saying how; the relief construction offered is stated not to be the tested one. · **Type:** Enablement · **Claims affected:** 16, 17, 26, 58, 65 · **Severity:** high · **Source:** F-28-3 · **How to respond:** As row 6; describe the prototype's outer-surface fit, clearance and closing line. · **Status:** addressed in fix pass (not re-reviewed)

#: 10 · **Deficiency or prior-art reference:** Obviousness: a staged-opening valve (US4044791A or US5918628A) combined with a ring deformed outward to open a gap (US5924444A) is a credible attack, and the application's own inspection found cured sealant on the surface the purge gap sweeps, undermining the effect relied on. · **Type:** Inventive step · **Claims affected:** 1, 2, 3, 38, 57 · **Severity:** high · **Source:** F-32-5 · **How to respond:** Add what the inventor recorded (state of the conventional cores' surfaces, identity of the comparison cores, evidence air passed the purge gap); restate the effect in [0012] and [0015] to what the data support. · **Status:** open, routed to fix pass

#: 11 · **Deficiency or prior-art reference:** Claim 3 claims the seat ring alone; everything beyond "a tube with an edge and a non-straight elastic wall" describes the valve core it is meant for. An examiner reads it on any suitable resilient corrugated tube. · **Type:** Novelty · **Claims affected:** 3, 4, 62–64, 70, 71, 74, 78, 82, 83 · **Severity:** high · **Source:** F-32-1 · **How to respond:** Human decision: recite ring structure a known tube would not have (e.g. the fit band and free portion of claim 66, or the bore-bounded gap of claim 65). Claims 1 and 77 keep the broad valve-core scope. · **Status:** addressed in fix pass (not re-reviewed)

#: 12 · **Deficiency or prior-art reference:** The description teaches an axially deflecting disc spring that opens a purge gap and says it is "not a radial expansion", so a competitor can copy it outside every claim; in the US it is dedicated to the public. · **Type:** Scope (design-around) · **Claims affected:** 1, 6 · **Severity:** high · **Source:** F-29-1 · **How to respond:** Add a claim to a "deflectable element" whose deflection opens the purge gap, with support statements and a figure. Nothing is narrowed. · **Status:** addressed in fix pass (not re-reviewed)

#: 13 · **Deficiency or prior-art reference:** Same escape seen from the claims: every independent claim requires radial expansion; the disc-spring form of [0181] is described, working and unclaimed. · **Type:** Scope (design-around) · **Claims affected:** 1, 2, 3 · **Severity:** high · **Source:** F-32-7 · **How to respond:** As row 12; add a figure of the disc-spring form and conform [0016]. · **Status:** open, routed to fix pass

#: 14 · **Deficiency or prior-art reference:** Same escape, with the drafting of a parallel valve-core and method claim to a "deflectable element". · **Type:** Scope (design-around) · **Claims affected:** 1, 2 · **Severity:** high · **Source:** F-33-2 · **How to respond:** As row 12. · **Status:** open, routed to fix pass

#: 15 · **Deficiency or prior-art reference:** A competitor lets the load-applying surface touch the element from lift-off and opens the purge gap only above a load threshold; there is then no "spaced … throughout the normal stroke" and no literal infringement. · **Type:** Scope (design-around) · **Claims affected:** 1, 2 · **Severity:** high · **Source:** F-33-3 · **How to respond:** Add a parallel claim bounding the travel by pressure thresholds and gap state, if the inventor conceived contact from lift-off. · **Status:** open, routed to fix pass

#: 16 · **Deficiency or prior-art reference:** "Closed at rest" is defined in [0069] so that any unloaded clearance counts as closed, so claim 1 does not exclude a permanent bypass that widens under load (e.g. US5924444A); FIG. 4A draws a clearance at rest. · **Type:** Clarity · **Claims affected:** 1, 2, 67 · **Severity:** high · **Source:** F-30-1 · **How to respond:** Redefine "closed at rest" functionally (contact, or no substantial flow at rest), with the inventor's confirmation; redraw FIG. 4A. · **Status:** addressed in fix pass (not re-reviewed)

#: 17 · **Deficiency or prior-art reference:** The express definitions of "seat ring" and "purge shoulder" are narrower than the claims (collar-loaded, taper-loaded and loose rings fall outside them), and a court applies an applicant's definitions. · **Type:** Clarity · **Claims affected:** 1, 15, 20 · **Severity:** high · **Source:** F-38-1 · **How to respond:** Reword [0060] and [0065] to the genus terms (load-applying surface; ring retained in use), and correct the locked terminology row. · **Status:** open, routed to fix pass

#: 18 · **Deficiency or prior-art reference:** Constructions the dossier records as not in the inventor's disclosure (slotted sleeve, split and tapered rings, bore relief, wall slots, FIG. 13A) are drawn and claimed; claims with no human inventor are exposed in US, UK and EPO. · **Type:** Support · **Claims affected:** 26, 40–44, 66, 67 · **Severity:** high · **Source:** F-30-7 · **How to respond:** Put each construction to the inventor; keep what is confirmed, delete the rest or record the real contributor; remove the provenance captions. · **Status:** addressed in fix pass (not re-reviewed)

#: 19 · **Deficiency or prior-art reference:** The figure files call those constructions unconfirmed candidates while the description and claims present them as embodiments; the package contradicts itself. · **Type:** Support · **Claims affected:** 40–44, 67 · **Severity:** high · **Source:** F-38-2 · **How to respond:** As row 18; reconcile once the inventor has decided. · **Status:** open, routed to fix pass

#: 20 · **Deficiency or prior-art reference:** Whether the inventor conceived those constructions is not recorded; unresolved, the claims reciting them face inventorship and entitlement challenges. · **Type:** Support · **Claims affected:** 26, 40–45, 67 · **Severity:** high · **Source:** F-36-1 · **How to respond:** Inventor confirms or declines each construction; unconfirmed forms come out of the claims. · **Status:** open

#: 21 · **Deficiency or prior-art reference:** US4044791A (City Tank Corp): floating seat biased against a poppet; further poppet travel unseats it and opens secondary passageways. Nearest structural analogue to over-travel plus purge gap; no cleaning purpose. Half of the obviousness combination. · **Type:** Prior art · **Claims affected:** 1, 2, 3, 5–10 · **Severity:** high · **Source:** US4044791A · **How to respond:** Distinguish on the deformable element whose shape recovery closes the gap, and on the rest-closed secondary path; strengthen the technical-effect evidence. · **Status:** prior art

#: 22 · **Deficiency or prior-art reference:** US5924444A (Innovent Inc): tyre-mounted relief valve whose ring is deformed radially outward to open a gap the closed valve does not have; elastomeric, flow-driven, deflation direction, no cleaning. The other half of the obviousness combination. · **Type:** Prior art · **Claims affected:** 1, 2, 3 · **Severity:** high · **Source:** US5924444A · **How to respond:** Rely on the load-applying surface moved by the poppet and a gap closed at rest; do not widen "closed at rest" beyond what separates this reference. · **Status:** prior art

#: 23 · **Deficiency or prior-art reference:** US5918628A (Parker Hannifin): spring-biased poppet whose further travel opens a larger second-stage flow path. Staged travel with a rigid seat; alternative first leg of the obviousness combination. · **Type:** Prior art · **Claims affected:** 1, 2, 5–10 · **Severity:** high · **Source:** US5918628A · **How to respond:** Distinguish on the deflecting element and the separate purge gap; the two-rate spring claims alone will not carry inventive step. · **Status:** prior art

#: 24 · **Deficiency or prior-art reference:** Claims 24 and 22 combined claim a single bow that the load flattens, but [0174] says the bow increases under load. · **Type:** Support · **Claims affected:** 22, 24 · **Severity:** medium · **Source:** F-28-8 · **How to respond:** Make claim 24 depend on claim 21 only, unless the geometry correction reconciles them. · **Status:** open

#: 25 · **Deficiency or prior-art reference:** The load-to-expansion mechanism is described two ways (convolution flattens; bow increases), and the 0.15 mm gap exceeds the 0.12 mm amplitude it is said to come from. · **Type:** Enablement · **Claims affected:** 22, 24, 64 · **Severity:** medium · **Source:** F-36-15 · **How to respond:** Add the at-rest and loaded wall profile of the prototype, from the inventor. · **Status:** open

#: 26 · **Deficiency or prior-art reference:** The "snap-back wipe" cannot happen in the stated order: the ring recovers while the sealing face is still a full normal stroke away; the mechanism is stated only as a belief. · **Type:** Enablement · **Claims affected:** 32, 56, 76 · **Severity:** medium · **Source:** F-28-19 · **How to respond:** State from the inventor what surface the recovering ring moves across, or recast the claims to shape recovery before the sealing face reseats. · **Status:** open

#: 27 · **Deficiency or prior-art reference:** Same wipe problem, found independently. · **Type:** Enablement · **Claims affected:** 32, 56, 76 · **Severity:** medium · **Source:** F-32-18 · **How to respond:** As row 26. · **Status:** open

#: 28 · **Deficiency or prior-art reference:** No test isolates the air purge or the wipe, and no cold or fatigue test exists; the only purge-gap observation is sealant still present there. · **Type:** Enablement · **Claims affected:** 1, 2,

3, 38, 57 · **Severity:** medium · **Source:** F-36-8 · **How to respond:** Before filing, test a core whose purge gap cannot open under the Example 1 protocol and report the result as found. · **Status:** open

#: 29 · **Deficiency or prior-art reference:** One species built (NiTi); polymer, spring-metal and other forms are claimed on expectation, and the text says it cannot give PEEK gap values. · **Type:** Enablement · **Claims affected:** 1, 30, 31, 40–43, 74, 75 · **Severity:** medium · **Source:** F-32-16 · **How to respond:** Add the inventor's design reasoning (e.g. strain in PEEK at the stated amplitude); keep the candour. · **Status:** open

#: 30 · **Deficiency or prior-art reference:** Genus breadth against one tested species, and no method of measuring crack pressure, over-travel differential, gap width or austenite finish temperature (EPO parameter clarity). · **Type:** Enablement · **Claims affected:** 1, 11, 19, 29–31, 33, 68, 74, 75 · **Severity:** medium · **Source:** F-30-11 · **How to respond:** State how each prototype value was measured or set; say where a value is a design value. · **Status:** open

#: 31 · **Deficiency or prior-art reference:** The actuating pressure depends on unrecorded pump flow, over-travel was inferred from an audible snap, and the claimed values cannot be reproduced from the text. · **Type:** Enablement · **Claims affected:** 5, 11, 12, 19, 29, 33, 34, 53, 72 · **Severity:** medium · **Source:** F-32-19 · **How to respond:** As row 30. · **Status:** open

#: 32 · **Deficiency or prior-art reference:** The evidence of over-travel (a snap on removing the pump head) does not fit the pressure mechanism described; it may show something else held the poppet open. · **Type:** Enablement · **Claims affected:** 5, 11, 12, 51–53 · **Severity:** medium · **Source:** F-36-6 · **How to respond:** Inventor states from the record when over-travel occurred and how the pump head engaged; correct [0143] and [0290] if needed. · **Status:** open

#: 33 · **Deficiency or prior-art reference:** Example 1 does not state the tested cores' convolution count, spring construction or purge-gap boundary, so a fallback to the tested embodiment risks intermediate generalisation. · **Type:** Support · **Claims affected:** 12, 34, 69 · **Severity:** medium · **Source:** F-29-4 · **How to respond:** Add those three facts for prototypes P-1 to P-3 from the inventor's records. · **Status:** open

#: 34 · **Deficiency or prior-art reference:** Figure captions for FIGS. 10–13A call the constructions candidates "shown for disclosure of the genus only", which an opponent will quote against possession. · **Type:** Support · **Claims affected:** 26, 40–44, 66, 67 · **Severity:** medium · **Source:** F-28-2 · **How to respond:** Remove the status wording from the drawn captions once the inventor has decided. · **Status:** open

#: 35 · **Deficiency or prior-art reference:** Slotted sleeve, split and tapered rings: no closed-at-rest purge gap and no seal at rest described, although claim 1 requires both. · **Type:** Support · **Claims affected:** 40–43 · **Severity:** medium · **Source:** F-28-17 · **How to respond:** For each form, state where the sealing edge is and what closes the purge gap at rest, if within the inventor's conception. · **Status:** open

#: 36 · **Deficiency or prior-art reference:** Two-piece seat: the wiper ring is not said to open a purge gap, numeral 146 is undescribed, and the purge shoulder's route past the rigid seat is missing. · **Type:** Support · **Claims affected:** 1 · **Severity:** medium · **Source:** F-28-20 · **How to respond:** Add to [0184] the location of purge gap 146, its closure at rest, its air route and the shoulder's path; inventor to confirm. · **Status:** open

#: 37 · **Deficiency or prior-art reference:** Same two-piece-seat gap, found independently. · **Type:** Support · **Claims affected:** 1 · **Severity:** medium · **Source:** F-32-22 · **How to respond:** As row 36. · **Status:** open

#: 38 · **Deficiency or prior-art reference:** The two-piece seat has no fallback claim; if claim 1 narrows towards the seat ring, it is lost. · **Type:** Support · **Claims affected:** 1, 20 · **Severity:** medium · **Source:** F-36-17 · **How to respond:** Add dependent claims on the rigid seat 140 and wiper ring 144; support exists at [0183]–[0185]. · **Status:** open

#: 39 · **Deficiency or prior-art reference:** [0037] and [0351] close the permitted combinations to a listed set, and several claimed combinations are not listed. · **Type:** Support · **Claims affected:** 19, 26, 30, 31, 36, 66 · **Severity:** medium · **Source:** F-28-11 · **How to respond:** Replace the closed wording with an open statement and list the claimed combinations. · **Status:** open

#: 40 · **Deficiency or prior-art reference:** Wall-slot arrangements are conflated, and the valve-stem-bore arrangement is one sentence with no figure; claim 47 omits claim 45. · **Type:** Support · **Claims affected:** 44, 45, 47 · **Severity:** medium · **Source:** F-29-8 · **How to respond:** Describe the two wall-slot arrangements separately; add a figure and paragraph for the valve-stem-bore form. · **Status:** open

#: 41 · **Deficiency or prior-art reference:** The seat-ring and purge-shoulder definitions contradict claims 3, 15 with 20, 43 and 77. · **Type:** Clarity · **Claims affected:** 3, 15, 20, 43, 77 · **Severity:** medium · **Source:** F-30-4 · **How to respond:** Reword [0060] to the load-applying surface and "retained, in use"; no claim change. · **Status:** open

#: 42 · **Deficiency or prior-art reference:** Same definitional conflict, found independently. · **Type:** Clarity · **Claims affected:** 3, 15, 20, 43, 77 · **Severity:** medium · **Source:** F-32-9 · **How to respond:** As row 41. · **Status:** open

#: 43 · **Deficiency or prior-art reference:** "Radially deflectable element", "return bias" and "load-converting geometry" are functional terms, and [0108] invites a US means-plus-function (112(f)) reading limited to disclosed structures. · **Type:** Clarity · **Claims affected:** 1, 3, 6 · **Severity:** medium · **Source:** F-30-18 · **How to respond:** Give each term structural content without narrowing the genus, and name the linked structures in [0108]; or counsel accepts the 112(f) reading. · **Status:** open

#: 44 · **Deficiency or prior-art reference:** Same 112(f) risk for "load-applying surface" and "radially deflectable element" in claim 1. · **Type:** Clarity · **Claims affected:** 1 · **Severity:** medium · **Source:** F-37-4 · **How to respond:** As row 43; a matter for counsel's claim-construction judgement. · **Status:** open

#: 45 · **Deficiency or prior-art reference:** Claim 45 bounds a valve core by a valve stem it does not contain; claim 49 contradicts claim 47; claim 50 defines a valve by the sealant in the tube. · **Type:** Formal · **Claims affected:** 45, 47, 49, 50 · **Severity:** medium · **Source:** F-30-13 · **How to respond:** Move claim 45 under claim 47; amend claim 47 to "received in, or formed in"; recast claim 50 as a tyre-chamber claim. · **Status:** open

#: 46 · **Deficiency or prior-art reference:** Claim 77 is a second independent valve-core claim of different scope (EPO Rule 43(2)); unity a posteriori and US restriction likely. · **Type:** Formal · **Claims affected:** 1, 2, 3, 4, 77 · **Severity:** medium · **Source:** F-30-20 · **How to respond:** Make claim 77 dependent or delete it; plan for a US divisional. · **Status:** open

#: 47 · **Deficiency or prior-art reference:** Claims fees and US multiple-dependency form: 86 claims attract excess fees in all three offices. · **Type:** Formal · **Claims affected:** 13, 16, 20, 33, 47, 61, 69, 77, 86 · **Severity:** medium · **Source:** F-30-23 · **How to respond:** Consolidate range and material fallbacks for EPO and UK; single-dependency US set. · **Status:** open

#: 48 · **Deficiency or prior-art reference:** FIG. 5 marks both pressure thresholds at the wrong end of their ranges and draws no preload, contradicting the definitions claims 5, 6 and 11 rest on. · **Type:** Formal · **Claims affected:** 5, 6, 11 · **Severity:** medium · **Source:** F-32-12 · **How to respond:** Redraw: curve starts at the crack-pressure force at zero travel; mark the over-travel differential at the change of slope. · **Status:** open

#: 49 · **Deficiency or prior-art reference:** Drawings: FIG. 13A and numeral 220 undescribed, numerals 402/404 unregistered, status captions on the sheets, and no figure of the collar, springs or valve-stem bore that are claimed (37 CFR 1.83(a)). · **Type:** Formal · **Claims affected:** 9, 10, 15, 26, 40–45, 67 · **Severity:** medium · **Source:** F-32-13 · **How to respond:** Describe or remove FIG. 13A and 220; register 402/404; add figures for claimed features; strip captions. · **Status:** open

#: 50 · **Deficiency or prior-art reference:** Drawings do not show claimed features (collar, spring constructions, valve-stem bore) and carry undescribed numerals. · **Type:** Formal · **Claims affected:** 9, 10, 15, 45 · **Severity:** medium · **Source:** F-30-15 · **How to respond:** As row 49. · **Status:** open

#: 51 · **Deficiency or prior-art reference:** FIG. 13A and numerals 146 and 220 have no description; claim 1 uses 146. · **Type:** Formal · **Claims affected:** 1 · **Severity:** medium · **Source:** F-28-13 · **How to respond:** Describe FIG. 13A and purge gap 146, or remove FIG. 13A and the numerals. · **Status:** open

#: 52 · **Deficiency or prior-art reference:** FIG. 13A (Schrader-pattern core body 220) is missing from the brief and detailed description. · **Type:** Formal · **Claims affected:** none · **Severity:** medium · **Source:** F-38-5 · **How to respond:** As row 51. · **Status:** open

#: 53 · **Deficiency or prior-art reference:** The thermal variant uses "purge gap" against its own definition, and numbered statements outside every claim remain in the description. · **Type:** Formal · **Claims affected:** none · **Severity:** medium · **Source:** F-32-21 · **How to respond:** Present the thermal variant as a separate aspect; recast the numbered statements as prose embodiments, keeping them as amendment basis. · **Status:** open

#: 54 · **Deficiency or prior-art reference:** The thermal variant is disclosed, unclaimed and excluded by the definitions; once published it is prior art against a later filing. · **Type:** Support · **Claims affected:** none · **Severity:** medium · **Source:** F-36-16 · **How to respond:** Human decision: remove it for a later filing, claim it with its own enabling text, or keep it as description only. · **Status:** open

#: 55 · **Deficiency or prior-art reference:** The description states the purge gap is never opened by pressure or flow; this disavows a pressure-expanded ring and forecloses a US equivalents case. · **Type:** Scope (design-around) · **Claims affected:** 1, 2, 3 · **Severity:** medium · **Source:** F-32-11 · **How to respond:** Recast [0018] and [0137] as statements about the embodiments described. · **Status:** open

#: 56 · **Deficiency or prior-art reference:** No independent claim covers cleaning by seat deflection and recovery without a secondary flow path, which is what the inspection result actually supports. · **Type:** Scope (design-around) · **Claims affected:** 1, 2, 3, 32, 56, 76 · **Severity:** medium · **Source:** F-33-4 · **How to respond:** Add an independent wipe claim; basis exists for the two-piece seat, the one-piece ring needs an inventor statement. · **Status:** open

#: 57 · **Deficiency or prior-art reference:** Every independent claim needs a contact-free first range of travel; a design with the surface resting on the element from the start escapes. Whether the inventor conceived that is not recorded. · **Type:** Scope (design-around) · **Claims affected:** 1, 2, 3 · **Severity:** medium · **Source:** F-32-3 · **How to respond:** If within conception, add the embodiment and a claim without the "spaced throughout" limitation. · **Status:** open

#: 58 · **Deficiency or prior-art reference:** A pressure-driven piston, the pressure itself, or a rigid sliding sleeve opens a second path outside the claims; whether the inventor conceived any is not recorded. · **Type:** Scope (design-around) · **Claims affected:** 1, 2, 3 · **Severity:** medium · **Source:** F-33-7 · **How to respond:** Ask the inventor; if conceived, add an embodiment and a dependent family. · **Status:** open

#: 59 · **Deficiency or prior-art reference:** Most fallbacks add conventional features (ordinary two-stage springs, NiTi valve seats, lock nut) or ranges with no stated effect, so if claim 1 falls they do not rescue it. · **Type:** Inventive step · **Claims affected:** 5, 8–11, 27–29, 35–37, 68, 81 · **Severity:** medium · **Source:** F-32-20 · **How to respond:** State the technical reason tying each key fallback to the purge-gap opening; add a mid-level claim combining claims 1, 20, 21 and 25. · **Status:** open

#: 60 · **Deficiency or prior-art reference:** No abstract in the package (required in all three offices). Raised at the 14-question gate; not separately adjudicated. · **Type:** Formal · **Claims affected:** 1, 2, 3 · **Severity:** medium · **Source:** F-gate14-1 · **How to respond:** Draft an abstract of at most 150 words on claims 1–3 in the locked terms and review it before Gate 3. · **Status:** open

#: 61 · **Deficiency or prior-art reference:** No drawing sheets exist, only figure instructions; late drawings risk missing-parts treatment and US new matter. Raised at the 14-question gate; not separately adjudicated. · **Type:** Formal · **Claims affected:** none · **Severity:** medium · **Source:** F-gate14-2 · **How to respond:** Render the sheets from the instructions, check numerals against the registry, file with the description. · **Status:** open

#: 62 · **Deficiency or prior-art reference:** US3473561A (Svensson): resilient closure member whose edge wipes the valving surfaces as it flexes, dislodging particles. Closest disclosure of a wipe from the closure element's own deformation. · **Type:** Prior art · **Claims affected:** 1, 32, 56, 76 · **Severity:** medium · **Source:** US3473561A · **How to respond:** Distinguish on over-travel loading of a separate element and the purge gap; the wipe claims alone are exposed. · **Status:** prior art

#: 63 · **Deficiency or prior-art reference:** US3570510A (Ishikawajima-Harima): clean air supplied through an annular slit to keep a valve seat clean. Air purge of a seat, but from a dedicated supply, not opened by the valve member. · **Type:** Prior art · **Claims affected:** 1, 2, 38, 57 · **Severity:** medium · **Source:** US3570510A · **How to respond:** Distinguish on the purge gap opened by the poppet's own over-travel. · **Status:** prior art

#: 64 · **Deficiency or prior-art reference:** EP3171056A1 (HS Wroclaw): shape-memory sealing ring with a wave-shaped section whose radial width changes on activation. Wave-section NiTi-type ring geometry, thermally driven static seal. · **Type:** Prior art · **Claims affected:** 3, 21–24, 27–29, 62–64 · **Severity:** medium · **Source:** EP3171056A1 · **How to respond:** Distinguish on radial expansion produced by axial load, not heat; relevant also to the unclaimed thermal variant. · **Status:** prior art

#: 65 · **Deficiency or prior-art reference:** US4050473A (Alfa-Wassermann): flexible disc on a valve head wipes dirt from the seat. Wipe on the valve member, not the seat. · **Type:** Prior art · **Claims affected:** 32, 56, 76 · **Severity:** medium · **Source:** US4050473A · **How to respond:** Distinguish on the seat element's recovery as the wiping action. · **Status:** prior art

#: 66 · **Deficiency or prior-art reference:** US6019118A (DaimlerChrysler): valve seat or tappet of superelastic NiTi, chosen to tolerate wear and contamination. Shows a NiTi valve seat in moving duty. · **Type:** Prior art · **Claims affected:** 27–29, 70–73 · **Severity:** medium · **Source:** US6019118A · **How to respond:** The NiTi claims will not carry inventive step on material alone; tie them to the recovery that closes the purge gap. · **Status:** prior art

#: 67 · **Deficiency or prior-art reference:** US11549591B1 (Technetics Group; continuations US11808363B1 and US12104701B1 carry the same description, not saved): poppet check valve with a resilient metal seat having a cylindrical sidewall and undulating flanges. No radial expansion, no second flow path. · **Type:** Prior art · **Claims affected:** 3, 21–24, 62–64 · **Severity:** medium · **Source:** US11549591B1 · **How to respond:** Key reference against claim 3; recite ring structure beyond an undulating wall. · **Status:** prior art

#: 68 · **Deficiency or prior-art reference:** Claim 4 claims shape-setting alone, while every method passage includes laser cutting and quenching. · **Type:** Support · **Claims affected:** 4 · **Severity:** low · **Source:** F-28-9 · **How to respond:** Add a sentence that the ring may be made by shape-setting with the other steps optional. · **Status:** open

#: 69 · **Deficiency or prior-art reference:** Ranges stated for the seat ring are claimed for every radially deflectable element. · **Type:** Support · **Claims affected:** 19, 33, 34 · **Severity:** low · **Source:** F-28-16 · **How to respond:** If the inventor agrees, state the ranges apply to every form; otherwise depend claims 19, 33, 34 on claim 20. · **Status:** open

#: 70 · **Deficiency or prior-art reference:** Purge-gap width range claimed for elements with no stated width. · **Type:** Support · **Claims affected:** 19 · **Severity:** low · **Source:** F-32-26 · **How to respond:** As row 69. · **Status:** open

#: 71 · **Deficiency or prior-art reference:** Normal stroke, interference fit, composition, shape-setting and austenite finish temperature each have one value, giving no room to amend. · **Type:** Support · **Claims affected:** 29, 34, 73, 81, 85 · **Severity:** low · **Source:** F-29-6 · **How to respond:** Ask the inventor for workable ranges and add them as candidates. · **Status:** open

#: 72 · **Deficiency or prior-art reference:** No range is stated for the normal stroke; a range cannot be added after filing. · **Type:** Support · **Claims affected:** 34 · **Severity:** low · **Source:** F-36-9 · **How to respond:** If the inventor has an operating window, state it with nested ranges containing 0.6 mm. · **Status:** open

#: 73 · **Deficiency or prior-art reference:** No dimensionless (ratio) fallback survives scaling of the core. · **Type:** Support · **Claims affected:** 11, 33, 53 · **Severity:** low · **Source:** F-36-19 · **How to respond:** If the inventor treats a ratio as the design rule, state it and its range before filing. · **Status:** open

#: 74 · **Deficiency or prior-art reference:** Mechanical actuation of the over-travel (pin, button) is implied as an alternative but not described. · **Type:** Support · **Claims affected:** 1, 2, 5, 51 · **Severity:** low · **Source:** F-36-7 · **How to respond:** If conceived, add a paragraph and a dependent method claim. · **Status:** open

#: 75 · **Deficiency or prior-art reference:** No valve pattern other than Presta is described, although FIG. 13A numbers a Schrader-pattern body. · **Type:** Support · **Claims affected:** 46, 47, 48 · **Severity:** low · **Source:** F-36-11 · **How to respond:** If conceived, describe the Schrader-pattern core with its dimensional differences. · **Status:** open

#: 76 · **Deficiency or prior-art reference:** Only prototype routes of manufacture are described; volume routes (moulding the polymer ring) are not. · **Type:** Support · **Claims affected:** 4, 78, 79, 80 · **Severity:** low · **Source:** F-36-21 · **How to respond:** Ask the inventor which volume routes they contemplate and add them. · **Status:** open

#: 77 · **Deficiency or prior-art reference:** The axially deflecting and tilting forms rest on one paragraph and no figure. · **Type:** Support · **Claims affected:** 1 · **Severity:** low · **Source:** F-36-14 · **How to respond:** Add a figure of the disc-spring form and a paragraph on its face, purge route and a tilting example. · **Status:** open

#: 78 · **Deficiency or prior-art reference:** Strut-pattern and helical-slot geometries have no drawing. · **Type:** Support · **Claims affected:** 3, 21 · **Severity:** low · **Source:** F-36-22 · **How to respond:** Add a two-panel figure. · **Status:** open

#: 79 · **Deficiency or prior-art reference:** Claim 1's numeral 146 has no counterpart in the description text. · **Type:** Support · **Claims affected:** 1 · **Severity:** low · **Source:** F-37-1 · **How to respond:** Add numeral 146 to [0184]. · **Status:** open

#: 80 · **Deficiency or prior-art reference:** Claim 9's spring alternatives omit the stop or staggered engagement that makes them two-rate. · **Type:** Support · **Claims affected:** 6, 9 · **Severity:** low · **Source:** F-37-2 · **How to respond:** Reinsert the wording of [0146] and [0147] into claim 9. · **Status:** open

#: 81 · **Deficiency or prior-art reference:** Claim 47 excludes the valve-stem-bore species of claim 45. · **Type:** Support · **Claims affected:** 45, 47 · **Severity:** low · **Source:** F-37-8 · **How to respond:** Extend claim 47's range to claims 45 and 46. · **Status:** open

#: 82 · **Deficiency or prior-art reference:** "Air" is not defined to include CO2 or nitrogen from a cartridge. · **Type:** Support · **Claims affected:** 2, 38, 57 · **Severity:** low · **Source:** F-36-18 · **How to respond:** Define air as the gas delivered by the pump head, including carbon dioxide and nitrogen. · **Status:** open

#: 83 · **Deficiency or prior-art reference:** Polymer seat ring and two-piece seat lack operative dimensions. · **Type:** Enablement · **Claims affected:** 30, 31, 74, 75 · **Severity:** low · **Source:** F-36-13 · **How to respond:** Add a polymer wall-thickness window and indicative two-piece dimensions if available; otherwise counsel weighs keeping the claims. · **Status:** open

#: 84 · **Deficiency or prior-art reference:** No value for cold-service or CO2 margin on the austenite finish temperature. · **Type:** Enablement · **Claims affected:** 29, 72 · **Severity:** low · **Source:** F-36-20 · **How to respond:** If the inventor has a target, state it with a range. · **Status:** open

#: 85 · **Deficiency or prior-art reference:** "Laser cutting to a profile of the convoluted wall" is unclear; the wave is pressed, not cut. · **Type:** Clarity · **Claims affected:** 79 · **Severity:** low · **Source:** F-28-10 · **How to respond:** State what "profile" means in [0255], or delete claim 79. · **Status:** open

#: 86 · **Deficiency or prior-art reference:** "About 0.03 mm" and "nominal composition" in claims draw an EPO clarity objection. · **Type:** Clarity · **Claims affected:** 73, 85 · **Severity:** low · **Source:** F-30-24 · **How to respond:** State a tolerance if the inventor has one, or drop the qualifiers in the European set. · **Status:** open

#: 87 · **Deficiency or prior-art reference:** "About" in claim 85 relies on an unquantified tolerance. · **Type:** Clarity · **Claims affected:** 85 · **Severity:** low · **Source:** F-37-9 · **How to respond:** As row 86. · **Status:** open

#: 88 · **Deficiency or prior-art reference:** Convolution amplitude is claimed where no convolution is required. · **Type:** Clarity · **Claims affected:** 68, 69 · **Severity:** low · **Source:** F-32-23 · **How to respond:** Restrict the amplitude alternative to claims 62–67. · **Status:** open

#: 89 · **Deficiency or prior-art reference:** FIG. 1 instructions place the main passage against the bore, blurring the distinction claim 1 rests on. · **Type:** Formal · **Claims affected:** 1, 2, 3 · **Severity:** low · **Source:** F-28-12 · **How**

to respond: Move the lead line for 26 to the path between sealing face 42 and sealing edge 32. · **Status:** open

#: 90 · **Deficiency or prior-art reference:** Figure files carry drafting-process statements; numerals 402 and 404 are undescribed. · **Type:** Formal · **Claims affected:** 1, 26, 40, 42–44 · **Severity:** low · **Source:** F-28-14 · **How to respond:** Remove process statements; describe or drop 402 and 404. · **Status:** open

#: 91 · **Deficiency or prior-art reference:** Numeral 146 missing from the description; text legends in captions. · **Type:** Formal · **Claims affected:** 1 · **Severity:** low · **Source:** F-30-16 · **How to respond:** Introduce 146 in [0183]–[0184]; reduce captions to view titles. · **Status:** open

#: 92 · **Deficiency or prior-art reference:** The thermal variant's use of "purge gap" conflicts with [0067]; unclaimed embodiments must be marked before EPO grant. · **Type:** Formal · **Claims affected:** 1, 2, 3, 4, 77 · **Severity:** low · **Source:** F-30-22 · **How to respond:** Word [0067] "in the first aspect"; plan divisionals if wanted. · **Status:** open

#: 93 · **Deficiency or prior-art reference:** Claim 77 re-introduces a valve core and poppet already introduced in claim 3; four claim categories invite US restriction. · **Type:** Formal · **Claims affected:** 1, 2, 3, 4, 47, 77 · **Severity:** low · **Source:** F-32-24 · **How to respond:** In claim 77, tie the poppet to the one seating on the ring; plan a seat-ring divisional. · **Status:** open

#: 94 · **Deficiency or prior-art reference:** "Split seat ring" in claim 42 collides with the locked synonym "split seat" for the two-piece seat; claim 50 limited by things outside the product. · **Type:** Formal · **Claims affected:** 13, 38, 42, 50, 61 · **Severity:** low · **Source:** F-32-25 · **How to respond:** Rename or declare "split seat ring"; recast claim 50 as a tyre-chamber claim. · **Status:** open

#: 95 · **Deficiency or prior-art reference:** The registry carries a stale note that the description and claims do not exist. · **Type:** Formal · **Claims affected:** none · **Severity:** low · **Source:** F-38-10 · **How to respond:** Remove the note. · **Status:** open

#: 96 · **Deficiency or prior-art reference:** Method claim 2 is performed by the cyclist; a manufacturer is at most an indirect infringer. · **Type:** Scope (design-around) · **Claims affected:** 2 · **Severity:** low · **Source:** F-32-14 · **How to respond:** No change; enforce through claims 1, 3 and 47. · **Status:** open

#: 97 · **Deficiency or prior-art reference:** "Load-converting geometry" has no strut-pattern or helical-slot dependent fallback. · **Type:** Scope (design-around) · **Claims affected:** 3, 21 · **Severity:** low · **Source:** F-33-5 · **How to respond:** Add dependent claims for the strut pattern ([0173]) and helical slot ([0175]). · **Status:** open

#: 98 · **Deficiency or prior-art reference:** Where the valve-stem bore bounds the purge gap, a core sold alone has no purge gap. · **Type:** Scope (design-around) · **Claims affected:** 1, 45, 46 · **Severity:** low · **Source:** F-33-6 · **How to respond:** Word claim 45 as "configured so that, when fitted in a valve stem, …". · **Status:** open

#: 99 · **Deficiency or prior-art reference:** Claim 3 is avoided by a part with no sealing edge (wiper ring) or no geometry of its own (ring spread by a cone). · **Type:** Scope (design-around) · **Claims affected:** 3 · **Severity:** low · **Source:** F-33-8 · **How to respond:** Add an independent claim to the deflectable element not limited to one carrying the sealing edge. · **Status:** open

#: 100 · **Deficiency or prior-art reference:** Claim 3 can only be proved through the core the ring is fitted to. · **Type:** Scope (design-around) · **Claims affected:** 3 · **Severity:** low · **Source:** F-33-11 · **How to respond:** No amendment; enforce spare rings through structural dependents 62–69 and 71–73. · **Status:** open

#: 101 · **Deficiency or prior-art reference:** Claim 4's shape-setting step leaves no mark in the product and is avoided by cutting already-superelastic tube. · **Type:** Scope (design-around) · **Claims affected:** 4, 78, 79, 80 · **Severity:** low · **Source:** F-33-12 · **How to respond:** Add a method claim forming the geometry by laser cutting; treat manufacture claims as fallbacks. · **Status:** open

#: 102 · **Deficiency or prior-art reference:** US20210323363A1 (Santa Cruz Bicycles): says Presta and Schrader valves on tubeless tyres are often clogged by sealant; answers with a larger bore. Establishes the problem in the field. · **Type:** Prior art · **Claims affected:** 1, 2, 3 · **Severity:** low · **Source:** US20210323363A1 · **How to respond:** Problem is known; rely on the mechanism, not the problem statement, for inventive step. · **Status:** prior art

#: 103 · **Deficiency or prior-art reference:** US11148485B2 (Orange Seal): tubeless valve stem whose plunger is used manually to clear sealant. Problem acknowledged, manual solution. · **Type:** Prior art · **Claims affected:** 1, 2, 3 · **Severity:** low · **Source:** US11148485B2 · **How to respond:** As row 102. · **Status:** prior art

#: 104 · **Deficiency or prior-art reference:** US1934572A (individual, 1933): conventional tyre valve core with a sealing element and spring-biased poppet threaded into a stem. The baseline architecture. · **Type:** Prior art · **Claims affected:** 1, 47 · **Severity:** low · **Source:** US1934572A · **How to respond:** Likely the examiner's starting point at the EPO; claim 1's characterising features are the over-travel, the element and the purge gap. · **Status:** prior art

#: 105 · **Deficiency or prior-art reference:** US3595265A (Caterpillar): groove in the valve face flushed clean by flow on opening. Flow-flushing of deposits at a seat. · **Type:** Prior art · **Claims affected:** 1, 2 · **Severity:** low · **Source:** US3595265A · **How to respond:** Distinguish on a separate rest-closed path opened by deformation. · **Status:** prior art

#: 106 · **Deficiency or prior-art reference:** US9611937B2 (Baker Hughes): superelastic Ni–Ti metal seals, prestrained or heat-treated. Material and processing of NiTi seals. · **Type:** Prior art · **Claims affected:** 4, 27–29, 70–73, 78–86 · **Severity:** low · **Source:** US9611937B2 · **How to respond:** Material and process claims need the purge-gap function to carry them. · **Status:** prior art

#: 107 · **Deficiency or prior-art reference:** US8100410B2 (Centro Ricerche Fiat): hollow superelastic ring whose reaction under axial compression maintains a seal. · **Type:** Prior art · **Claims affected:** 27–29, 70–73 · **Severity:** low · **Source:** US8100410B2 · **How to respond:** As row 106. · **Status:** prior art

#: 108 · **Deficiency or prior-art reference:** US20040194970A1 (individual): folded seal expanded radially by embedded SMA wire; downhole. · **Type:** Prior art · **Claims affected:** 3, 21–24, 27 · **Severity:** low · **Source:** US20040194970A1 · **How to respond:** Distinguish on axial-load conversion by the wall itself. · **Status:** prior art

#: 109 · **Deficiency or prior-art reference:** US9126251B2 (Abbott Cardiovascular): laser cutting NiTi tube, oxide removal, heat treatment, electropolishing; the standard stent process chain. · **Type:** Prior art · **Claims affected:** 4, 78–86 · **Severity:** low · **Source:** US9126251B2 · **How to respond:** Manufacture claims are conventional steps; keep them as fallbacks only. · **Status:** prior art

#: 110 · **Deficiency or prior-art reference:** US11179979B2 (Sport Components AG): self-sealing one-way valve at the tyre end of the stem that keeps sealant out of the core. A tool-free answer to fouling by exclusion. · **Type:** Prior art · **Claims affected:** 1, 47–50 · **Severity:** low · **Source:** US11179979B2 · **How to respond:** Background to the problem; different solution. · **Status:** prior art

#: 111 · **Deficiency or prior-art reference:** US20050210675A1 (individual): removable valve core used to add sealant through the valve. Background practice. · **Type:** Prior art · **Claims affected:** 47–50 · **Severity:** low · **Source:** US20050210675A1 · **How to respond:** Background only. · **Status:** prior art

#: 112 · **Deficiency or prior-art reference:** Requester-supplied S1, marked "FICTIONAL US 2019/0123456 A1": a test reference, not a real publication; not searched or fetched. Its number is cited in [0009]. · **Type:** Prior art · **Claims affected:** none · **Severity:** low · **Source:** US 2019/0123456 A1 (S1) · **How to respond:** Not prior art; remove the citation from the description. · **Status:** prior art

#: 113 · **Deficiency or prior-art reference:** Requester-supplied S2, marked "FICTIONAL EP 3 456 789 B1": test reference with a seat lip deflecting outward under inflation and an elastic return. Not a real publication. · **Type:** Prior art · **Claims affected:** none · **Severity:** low · **Source:** EP 3 456 789 B1 (S2) · **How to respond:** Not prior art; if a real document like it exists, the professional search should find it. · **Status:** prior art

#: CN-1 · **Deficiency or prior-art reference:** Claim 3: the purge interval limitation needs a disclosed range. · **Type:** support (claims need disclosure) · **Claims affected:** 3 · **Severity:** medium · **Source:** claims fix pass · **How to respond:** add supporting disclosure before filing, or amend the claim · **Status:** open

How the score was reached

- **Novelty — 2 / 2.** No saved reference discloses every feature of claim 1 or claim 3; none of the 19 saved references scored above 3 of 5 for closeness, and for claim 3 the ruling found no saved reference that turns axial load into radial expansion.
- **Inventive step — 1 / 2.** A credible combination exists (US4044791A or US5918628A with US5924444A), and the application's own inspection does not show the purge-gap effect it relies on (F-32-5, upheld high).
- **Support and enablement — 1 / 2.** No critical finding, but open high findings: the Example 1 dates and cycle counts are inconsistent, the tested construction's geometry is not described coherently, and unconfirmed constructions are claimed.
- **Clarity and form — 1 / 2.** No critical finding, but open high findings: the fictional Background citation, the US multiple-dependency form, the empty definition of "closed at rest" and definitions narrower than the claims.
- **Eligibility and fallbacks — 1 / 2.** A mechanical valve core is eligible in the US, at the EPO and in the UK. The fallback ladder exists but is thin where it matters: most rungs add conventional features or untested ranges, and the disclosed disc-spring and contact-from-lift-off forms escape every claim.

Total 6 / 10. The US, EPO and UK positions do not differ materially: the US adds the multiple-dependency, means-plus-function and inventorship points, and the EPO and UK add intermediate generalisation and clarity, but none moves a factor. The score rests only on the prior art we found and saved. The search was thin (no Espacenet, CNIPA, DEPATISnet or J-PlatPat; no product literature), so a professional search, which the index recommends before filing, could lower it.

AI-assisted estimate for review by a qualified patent attorney or agent. Not legal advice and not a patentability opinion. Based only on the prior art identified in this matter.